

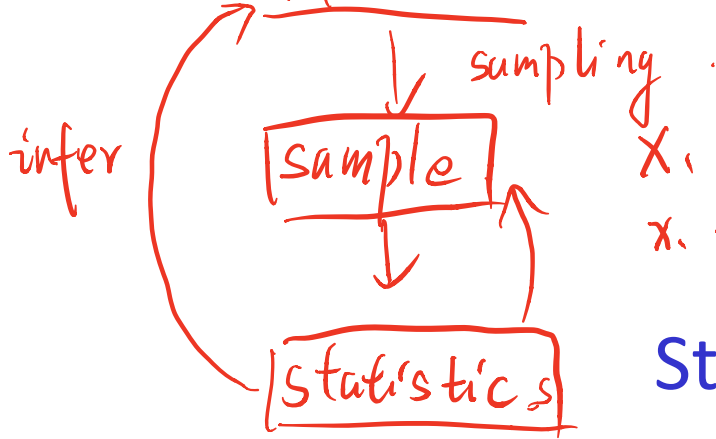
Statistical Modelling and Inference

Week2 estimation

Yu Zhang

population 总体

$E(X)$ $Var(X)$ 25%



$X_1 \dots X_n$

$x_1 \dots x_n \rightarrow$ sample value.

Statistics

a function of $T(X_1, \dots, X_n)$
data.

Setup

Suppose $Y_1, Y_2, \dots, Y_n$ are random variables with CDF F_θ , for some $\theta \in \Theta$, where Θ denotes the set of legitimate parameter values, called *parameter space*.

Parameter are descriptive measure of *entire population*

e.g. $Y_1, Y_2, \dots, Y_n \sim N(\mu, \sigma^2)$

parameter space : $\Theta = \{(\mu, \sigma^2) \mid \mu \in \mathbb{R}, \sigma^2 > 0\}$

parameters $\theta = \{\mu, \sigma\}$

We would like to estimate the unknown parameter θ from the data $y_1, y_2, \dots, y_n$.

Definition 1.1

A function $T(Y_1, Y_2, \dots, Y_n)$ is called a *statistic*.

A statistic T that takes values in Θ is called an *estimator* for θ .

~~A statistic is any function of data~~

$$T = \sum_{i=1}^n Y_i \quad \text{is a statistic.}$$

$$T = \min \{Y_1, Y_2, \dots, Y_m\} \quad \text{is a statistic}$$

$$T = Y_1 \quad \dots \quad \dots$$

An estimator is a statistic that is used to infer unknown parameter θ .

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \quad \text{is an estimator of } \mu$$

$$Y_1 \quad \dots \quad \dots \quad \dots \quad \dots \quad \dots$$

Bias

Definition 1.2

Let T be an estimator for θ , the *bias* of T is defined by

$$b_T(\theta) = E[T] - \theta.$$

If $b_T(\theta) = 0$ for all θ , then T is said to be an *unbiased estimator* for θ .

Review of $E(X)$ and $\text{Var}(X)$

$$E(X) = \sum_{i=1}^{\infty} x_i p(x_i) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E(c) = c$$

$$E(E(X)) = E(X)$$

$$E(cX) = c E(X)$$

X, Y independent
 $\Downarrow \Uparrow X$

$$E(X \pm Y) = E(X) \pm E(Y)$$

$$E(XY) = E(X)E(Y)$$

$$\text{Var}(X) = E(X - E(X))^2 = E(X^2) - E^2(X)$$

$$\text{Var}(c) = 0$$

$$\text{Var}(cX) = c^2 \text{Var}(X)$$

$$\text{Var}(aX + c) = a^2 \text{Var}(X)$$

X, Y independent

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) = 0$$

Example 1.1

同是同分布

Suppose $Y_1, Y_2, \dots, Y_n$ are independent and identically distributed (IID) $N(\mu, \sigma^2)$ random variables and let

$$E(Y_i) = \mu$$

$$\text{Var}(Y_i) = \sigma^2$$

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

be an estimator for μ . Calculate $b_{\bar{Y}}(\mu)$.

Example 1.1 Solutions

$$\begin{aligned}b_{\bar{Y}}(u) &= E(\bar{Y}) - u \\&= E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) - u \\&= \frac{1}{n} \sum_{i=1}^n E(Y_i) - u \\&= \frac{1}{n} \cdot n u - u \\&= 0\end{aligned}$$

$\bar{Y}$ is an unbiased estimator of u

Example 1.2

Suppose $Y_1, Y_2, \dots, Y_n$ are IID Bernoulli random variables with probability of success θ and let

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

$$E(Y_i) = \theta$$

$$\text{Var}(Y_i) = \theta(1-\theta)$$

be an estimator for θ . Calculate $b_{\bar{Y}}(\theta)$.

$$Y_1, \dots, \theta$$

Example 1.2 Solutions

$$\begin{aligned}b_{\bar{Y}}(\theta) &= E(\bar{Y}) - \theta \\&= \frac{1}{n} \sum E(Y_i) - \theta \\&= \frac{1}{n} \cdot n \theta - \theta \\&= 0\end{aligned}$$

$$\begin{aligned}b_{Y_1}(\theta) &= E(Y_1) - \theta \\&= \theta - \theta \\&= 0\end{aligned}$$



Definition 1.3

Let T be an estimator for θ , the *mean squared error* (MSE) of T is defined by

$$MSE_T(\theta) = E[(T - \theta)^2].$$

Example 1.3

Suppose $Y_1, Y_2, \dots, Y_n$ are IID $N(\mu, \sigma^2)$ random variables and let

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

be an estimator for μ .

Calculate $MSE_{\bar{Y}}(\mu)$.

Which estimator has a lower MSE, Y_1 or $\bar{Y}$?

$$MSE_{Y_1}(\mu) = E \{ Y_1 - E(Y_1) \}^2 = \text{Var}(Y_1) = 6^2$$

Example 1.3 Solutions u

$$\begin{aligned} \text{MSE}_{\bar{Y}}(u) &= E\left[\frac{(\bar{Y} - E(\bar{Y}))^2}{1}\right] \\ &= \text{Var}(\bar{Y}) \\ &= \text{Var}\left(\frac{1}{n} \sum Y_i\right) \\ &= \frac{1}{n^2} \text{Var}\left(\sum Y_i\right) \\ &= \frac{1}{n^2} n \text{Var}(Y_i) \\ &= \frac{\text{Var}(Y_i)}{n} = \frac{6^2}{n} \end{aligned}$$

Theorem 1

$$\text{Var}(X) = \underline{E(X^2)} - E^2(X)$$

$$MSE_T(\theta) = \text{var}(T) + b_T(\theta)^2.$$

RV

$$\begin{aligned} MSE_T(\theta) &= E\left\{(\underline{T} - \theta)^2\right\} \\ &= \text{Var}(T - \theta) + \left[E(T - \theta)\right]^2 \\ &= \text{Var}(T) + b_T^2(\theta) \end{aligned}$$

Proof of Theorem

$$MSE_T(\theta) = E(T - \theta)^2$$

$$= E \left\{ \underbrace{T - E(T)}_a + \underbrace{(E(T) - \theta)}_b \right\}^2$$

$$= E \left\{ (T - E(T))^2 \right\} - 2 E \left\{ (T - E(T))(E(T) - \theta) \right\} + E \left\{ (E(T) - \theta)^2 \right\}$$

$\nearrow E(T) - E(E(T)) = 0$

$$= Var(T) - 2(E(T) - \theta)E(T - E(T))$$

$$+ b_T(\theta)^2$$

Example 1.4

Suppose $Y_1, Y_2, \dots, Y_n$ are IID Bernoulli random variables with probability of success θ and let

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

be an estimator for θ .

Prove that $MSE_{\bar{Y}}(\theta) = \frac{\theta(1-\theta)}{n}$.

Example 1.4 Solutions

$$\begin{aligned} \text{MSE}_{\bar{y}}(\theta) &= \text{Var}(\bar{Y}) + \underbrace{b_{\bar{Y}}^2(\theta)}_{=0} \\ &= \frac{1}{n^2} \text{Var}(\sum Y_i) \end{aligned}$$

$$= \frac{1}{n^2} n \text{Var}(Y_i)$$

$$= \frac{1}{n^2} n \cdot \theta(1-\theta)$$

$$= \frac{\theta(1-\theta)}{n}$$

Linear estimator

Definition 1.4

An estimator of the form:

$$T = a_1 Y_1 + a_2 Y_2 + \cdots + a_n Y_n = \sum_{i=1}^n a_i Y_i,$$

for some constants $a_1, a_2, \dots, a_n$ is called a **linear estimator**.

$$T = \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \quad \text{is a linear estimator.}$$

$$T = Y_1$$

$$T = Y_1^2$$

is not a linear estimator.

Example 1.5

Is the sample mean $\bar{Y}$ a linear estimator?

Lemma 1

Suppose $Y_1, Y_2, \dots, Y_n$ are independent random variables with
 $E[Y_i] = \mu_i$ and $\text{var}(Y_i) = \sigma_i^2$.

Let

$$T = \sum_{i=1}^n a_i Y_i,$$

then

$$E[T] = ? \sum_{i=1}^n a_i \mu_i \text{ and}$$

$$\text{var}(T) = ? \sum_{i=1}^n a_i^2 \sigma_i^2$$

$$E(T) = E\left(\sum_{i=1}^n a_i Y_i\right) = \sum_{i=1}^n a_i E(Y_i) = \sum_{i=1}^n a_i \mu_i$$

$$\text{Var}(T) = \text{Var}\left(\sum_{i=1}^n a_i Y_i\right) = \sum_{i=1}^n a_i^2 \sigma_i^2$$

Furthermore, if

$Y_i \sim N(\mu_i, \sigma_i^2)$ independently, then

$$T \sim N\left(\sum_{i=1}^n a_i \mu_i, \sum_{i=1}^n a_i^2 \sigma_i^2\right).$$

BLUE

Definition 1.5

The *best Linear Unbiased Estimator* (BLUE) for a parameter θ is the estimator for θ linear, unbiased that has minimum variance.

- ① unbiased estimator
- ② Linear estimator
- ③ best: T has min Variance

Lemma 2

Suppose $Y_1, Y_2, \dots, Y_n$ are IID random variables with
 $E[Y_i] = \mu$ and $\text{var}(Y_i) = \sigma^2$.

The linear estimator

$$T = \sum_{i=1}^n a_i Y_i$$

is unbiased for μ if and only if

$$\sum_{i=1}^n a_i = 1.$$

iff

Proof of Lemma 2

$$\begin{aligned} E(T) &= E\left(\sum_{i=1}^n a_i Y_i\right) = E(a_1 Y_1) + E(a_2 Y_2) + \dots + E(a_n Y_n) \\ &= \sum_{i=1}^n a_i E(Y_i) \\ &= \sum_{i=1}^n a_i \mu \end{aligned}$$

Theorem 3

Suppose $Y_1, Y_2, \dots, Y_n$ are IID random variables with
 $E[Y_i] = \mu$ and $\text{var}(Y_i) = \sigma^2$,

then the BLUE for μ is given by

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i.$$

- ① $E(\bar{Y}) = \mu$ unbiased estimator
- ② $\bar{Y}$ is linear estimator
- ③ $\bar{Y}$ has min variance

Proof of Theorem 3

$$\begin{aligned}
 \text{Var} \left(\sum_{i=1}^n a_i y_i \right) &= \sum_{i=1}^n a_i^2 \sigma^2 \\
 &= \sigma^2 \sum_{i=1}^n a_i^2 \\
 &= \sigma^2 \sum_{i=1}^n \left(a_i - \frac{1}{n} + \frac{1}{n} \right)^2 \\
 &= \sigma^2 \sum_{i=1}^n \left\{ \underbrace{\left(a_i - \frac{1}{n} \right)^2}_a + \frac{2}{n} \left(a_i - \frac{1}{n} \right) + \frac{1}{n^2} \right\} \\
 &= \sigma^2 \left(\sum_{i=1}^n \left(a_i - \frac{1}{n} \right)^2 + \frac{2}{n} \sum_{i=1}^n \left(a_i - \frac{1}{n} \right) + \frac{1}{n} \right) \\
 &= \sigma^2 \left(\sum_{i=1}^n \left(a_i - \frac{1}{n} \right)^2 + \frac{1}{n} \right) \\
 &= \sigma^2 \sum_{i=1}^n \left(a_i - \frac{1}{n} \right)^2 + \frac{\sigma^2}{n} = \text{Var}(\bar{Y})
 \end{aligned}$$

Proof of Theorem 3 (cont.)

Definition 1.6

If T is an unbiased estimator for θ , then the standard deviation of the estimator is called the **standard error**:

$$SE(T) = \sqrt{\text{var}(T)}.$$